

Estimating the Carryover Parameter r and Dose-Specific Toxicity Probabilities p_k

1 Notation and observed-data model

Consider K dose levels. At dose k , let

$$n_{R,k}, \quad y_{R,k}$$

denote the number of regular observations and the number of regular DLTs, and let

$$n_{I,k}, \quad y_{I,k}$$

denote the number of intra-patient dose escalation (IPDE) observations and the number of IPDE DLTs.

Let p_k be the true current-dose toxicity probability at dose k . Let r be a common carryover probability shared across all dose levels. The model is

$$\begin{aligned} Y_{R,k} &\sim \text{Binomial}(n_{R,k}, p_k), \\ Y_{I,k} &\sim \text{Binomial}\{n_{I,k}, q_k\}, \quad q_k = r + (1 - r)p_k. \end{aligned}$$

The interpretation is that regular observations have toxicity probability p_k , whereas IPDE observations have an inflated toxicity probability q_k due to possible carryover:

$$q_k = r + (1 - r)p_k.$$

At the population level, if r and q_k are known, then

$$p_k = \frac{q_k - r}{1 - r}, \quad r < 1.$$

The full observed-data likelihood is

$$\begin{aligned} L(p_1, \dots, p_K, r) &= \prod_{k=1}^K p_k^{y_{R,k}} (1 - p_k)^{n_{R,k} - y_{R,k}} \\ &\quad \times \{r + (1 - r)p_k\}^{y_{I,k}} \{(1 - r)(1 - p_k)\}^{n_{I,k} - y_{I,k}}. \end{aligned}$$

The common parameter r links information across all dose levels, while each p_k remains dose-specific.

2 Estimation of p_k when r is provided

This section describes two estimators of p_k conditional on a fixed value of r . These are the estimators into which a posterior estimate of r will later be plugged.

2.1 Pooled moment estimator

The regular observations estimate p_k directly:

$$\hat{p}_{R,k} = \frac{y_{R,k}}{n_{R,k}}.$$

The IPDE observations estimate

$$q_k = r + (1 - r)p_k.$$

Thus, if r is fixed, the IPDE-based estimate of p_k is

$$\hat{p}_{I,k}(r) = \frac{y_{I,k}/n_{I,k} - r}{1 - r}.$$

The pooled moment estimator combines the regular and adjusted IPDE estimates by sample-size weighting:

$$\hat{p}_{k,\text{pool}}(r) = \frac{n_{R,k}\hat{p}_{R,k} + n_{I,k}\hat{p}_{I,k}(r)}{n_{R,k} + n_{I,k}}.$$

Equivalently,

$$\hat{p}_{k,\text{pool}}(r) = \frac{y_{R,k} + n_{I,k} \left(\frac{y_{I,k}/n_{I,k} - r}{1 - r} \right)}{n_{R,k} + n_{I,k}}.$$

In implementation, the estimator should be truncated to the unit interval:

$$\hat{p}_{k,\text{pool}}(r) = \text{clamp} \left(\frac{y_{R,k} + n_{I,k} \left(\frac{y_{I,k}/n_{I,k} - r}{1 - r} \right)}{n_{R,k} + n_{I,k}}, 0, 1 \right),$$

where

$$\text{clamp}(x, 0, 1) = \min\{\max(x, 0), 1\}.$$

This estimator is simple and intuitive, but it is not the exact likelihood maximizer in general. It treats the regular estimate and the adjusted IPDE estimate as two separate moment estimators and then pools them.

2.2 Exact conditional MLE

For a fixed value of r , the dose-specific likelihood contribution is

$$\begin{aligned} L_k(p_k; r) = & p_k^{y_{R,k}} (1 - p_k)^{n_{R,k} - y_{R,k}} \\ & \times \{r + (1 - r)p_k\}^{y_{I,k}} \{(1 - r)(1 - p_k)\}^{n_{I,k} - y_{I,k}}. \end{aligned}$$

Ignoring constants that do not involve p_k , the log-likelihood is

$$\ell_k(p_k; r) = y_{R,k} \log p_k + B_k \log(1 - p_k) + y_{I,k} \log\{r + (1 - r)p_k\},$$

where

$$B_k = n_{R,k} - y_{R,k} + n_{I,k} - y_{I,k}.$$

The score equation is

$$\frac{y_{R,k}}{p_k} - \frac{B_k}{1 - p_k} + \frac{y_{I,k}(1 - r)}{r + (1 - r)p_k} = 0.$$

Define

$$N_k = n_{R,k} + n_{I,k},$$

and

$$b_k(r) = y_{R,k}(1 - 2r) - B_k r + y_{I,k}(1 - r).$$

For $0 \leq r < 1$ and $N_k > 0$, the exact conditional MLE is the positive root

$$\hat{p}_{k,\text{mle}}(r) = \frac{b_k(r) + \sqrt{b_k(r)^2 + 4(1 - r)N_k y_{R,k} r}}{2(1 - r)N_k},$$

with truncation to $[0, 1]$ if needed:

$$\hat{p}_{k,\text{mle}}(r) = \text{clamp} \left(\frac{b_k(r) + \sqrt{b_k(r)^2 + 4(1 - r)N_k y_{R,k} r}}{2(1 - r)N_k}, 0, 1 \right).$$

This estimator uses the exact observed-data likelihood at dose k conditional on r .

2.3 Special cases

If $r = 0$, then IPDE observations have the same toxicity probability as regular observations. Both estimators reduce to the ordinary pooled binomial estimator:

$$\hat{p}_k = \frac{y_{R,k} + y_{I,k}}{n_{R,k} + n_{I,k}}.$$

If there are only IPDE observations at dose k , then the moment estimator and the conditional MLE reduce to

$$\hat{p}_k(r) = \left[\frac{y_{I,k}/n_{I,k} - r}{1 - r} \right]_0^1.$$

If there are only regular observations at dose k , then the conditional MLE reduces to

$$\hat{p}_k = \frac{y_{R,k}}{n_{R,k}}.$$

3 Posterior estimation of the common carryover parameter r

Because r is unknown and shared across all dose levels, it can be estimated using the data accumulated from all dose levels.

Assign a prior to r , for example

$$r \sim \text{Beta}(a_r, b_r),$$

possibly truncated to a clinically plausible interval $[0, r_{\max}]$, such as $[0, 0.4]$.

A useful default is

$$r \sim \text{Beta}(1, 9),$$

which has prior mean 0.10. Alternative sensitivity priors include

$$\text{Beta}(1, 4), \quad \text{Beta}(2, 8), \quad \text{Beta}(1, 1).$$

To obtain the posterior distribution of r , the nuisance parameters $p_1, \dots, p_K$ can be integrated out. Assign independent priors

$$p_k \sim \text{Beta}(a_p, b_p), \quad k = 1, \dots, K.$$

Then the marginal likelihood contribution for dose k is

$$\begin{aligned} m_k(r) = & \int_0^1 p_k^{y_{R,k}} (1 - p_k)^{n_{R,k} - y_{R,k}} \{r + (1 - r)p_k\}^{y_{I,k}} \\ & \times \{(1 - r)(1 - p_k)\}^{n_{I,k} - y_{I,k}} \pi(p_k) dp_k. \end{aligned}$$

The posterior distribution of the global carryover parameter is

$$\pi(r \mid D) \propto \pi(r) \prod_{k=1}^K m_k(r), \quad 0 \leq r \leq r_{\max}.$$

This posterior uses data from all dose levels through the product

$$\prod_{k=1}^K m_k(r).$$

Therefore, r is not estimated separately at each dose. Instead, a single global carryover parameter is learned from the entire accumulated dataset.

3.1 Grid implementation

A simple computational approach is to use a grid

$$0 = r_1 < r_2 < \dots < r_G = r_{\max}.$$

For each grid value r_g :

1. Compute $m_k(r_g)$ for each dose k by one-dimensional numerical integration over p_k .

2. Compute the unnormalized posterior weight

$$w_g = \pi(r_g) \prod_{k=1}^K m_k(r_g).$$

3. Normalize the weights:

$$\tilde{w}_g = \frac{w_g}{\sum_{h=1}^G w_h}.$$

The posterior mean estimate is

$$\hat{r}_{\text{post}} = \text{E}(r \mid D) \approx \sum_{g=1}^G r_g \tilde{w}_g.$$

Alternatively, one may use the posterior mode:

$$\hat{r}_{\text{MAP}} = \arg \max_{r_g} \tilde{w}_g.$$

For a plug-in approach, either $\hat{r}_{\text{post}}$ or $\hat{r}_{\text{MAP}}$ can be used. The posterior mean is usually smoother, while the posterior mode is closer to a penalized likelihood estimate.

3.2 Profile-posterior approximation

As a faster approximation, one may replace the integrated marginal likelihood $m_k(r)$ by the profile likelihood

$$L_k\{\hat{p}_{k,\text{mle}}(r); r\}.$$

This gives

$$\pi_{\text{prof}}(r \mid D) \propto \pi(r) \prod_{k=1}^K L_k\{\hat{p}_{k,\text{mle}}(r); r\}.$$

The fully Bayesian version integrates over p_k ; the profile-posterior version plugs in the conditional MLE of p_k for each fixed r .

4 Plugging the posterior estimate of r into the p_k estimators

After estimating the global carryover parameter using data from all dose levels, plug the posterior estimate of r into the dose-specific estimators from Section 2.

Let

$$\hat{r} = \hat{r}_{\text{post}}$$

be the posterior mean, or let

$$\hat{r} = \hat{r}_{\text{MAP}}$$

be the posterior mode.

4.1 Plug-in pooled estimator

The plug-in pooled estimator is

$$\hat{p}_{k,\text{pool}} = \hat{p}_{k,\text{pool}}(\hat{r}),$$

where

$$\hat{p}_{k,\text{pool}}(\hat{r}) = \text{clamp} \left(\frac{y_{R,k} + n_{I,k} \left(\frac{y_{I,k}/n_{I,k} - \hat{r}}{1 - \hat{r}} \right)}{n_{R,k} + n_{I,k}}, 0, 1 \right).$$

This corresponds to first estimating the common carryover strength from all dose levels, then adjusting the IPDE toxicity rate at each dose using the estimated carryover probability.

4.2 Plug-in exact-MLE estimator

The plug-in exact-MLE estimator is

$$\hat{p}_{k,\text{mle}} = \hat{p}_{k,\text{mle}}(\hat{r}),$$

where

$$\hat{p}_{k,\text{mle}}(\hat{r}) = \text{clamp} \left(\frac{b_k(\hat{r}) + \sqrt{b_k(\hat{r})^2 + 4(1 - \hat{r})N_k y_{R,k} \hat{r}}}{2(1 - \hat{r})N_k}, 0, 1 \right),$$

with

$$b_k(\hat{r}) = y_{R,k}(1 - 2\hat{r}) - B_k \hat{r} + y_{I,k}(1 - \hat{r}),$$

$$B_k = n_{R,k} - y_{R,k} + n_{I,k} - y_{I,k}, \quad N_k = n_{R,k} + n_{I,k}.$$

This estimator uses the exact likelihood contribution at dose k conditional on the globally estimated $\hat{r}$.

5 Summary algorithm

The proposed procedure can be summarized as follows.

Step 1. At each dose k , collect

$$(y_{R,k}, n_{R,k}, y_{I,k}, n_{I,k}).$$

Step 2. Specify priors

$$r \sim \text{Beta}(a_r, b_r), \quad p_k \sim \text{Beta}(a_p, b_p).$$

Step 3. Estimate the posterior distribution of r using all dose levels:

$$\pi(r \mid D) \propto \pi(r) \prod_{k=1}^K m_k(r).$$

Step 4. Compute a plug-in estimate of r , for example

$$\hat{r} = \text{E}(r \mid D).$$

Step 5. Estimate each dose-specific toxicity probability using either

$$\hat{p}_{k,\text{pool}} = \hat{p}_{k,\text{pool}}(\hat{r})$$

or

$$\hat{p}_{k,\text{mle}} = \hat{p}_{k,\text{mle}}(\hat{r}).$$

In words, the estimator first learns the common carryover parameter r from all dose levels and then uses this estimated carryover strength to recover the dose-specific current-toxicity probabilities p_k .

6 Remarks

The plug-in approach is simple and compatible with CFO-style decision rules because it produces a dose-specific toxicity estimate $\hat{p}_k$ for each dose. However, it ignores posterior uncertainty in r when estimating p_k . A fully Bayesian version would instead compute

$$\hat{p}_k = \text{E}(p_k \mid D) = \int \text{E}(p_k \mid r, D) \pi(r \mid D) dr.$$

The plug-in version is useful for implementation because it separates the procedure into two transparent parts: global estimation of r and dose-level estimation of p_k . This also makes it easy to compare the pooled estimator and the exact conditional MLE estimator in simulation.